

Charles Cleivin

Senior Software Engineer – AI, LLM Fine-tuning, Agent Systems, Rust & React
Canada / Brazil / USA (Remote or relocation. Available now)

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Professional Summary

Innovative and hands-on Senior Software Engineer with 10+ years of experience building distributed systems, intelligent agent platforms, and production-grade cross-platform mobile applications. AI/ML Specialist with deep expertise in LLM fine-tuning, cloud GPU infrastructure (H100/A100/RTX 4090), real-time computer vision, and end-to-end ML system design. Proven track record fine-tuning large language models for production content generation at scale. Creator of fully offline instant AI agents and real-time inference tools deployable in air-gapped environments. Strong knowledge of scalable backend APIs, full-stack tools like Next.js, and deep technical experience from low-level AI frameworks to high-level UX.

Core Technical Skills

LLM & Fine-tuning: Production experience fine-tuning 7B+ parameter models (Qwen2.5, Llama) using LoRA, 4-bit quantization (NF4), and PEFT. Expertise in memory optimization, smart chunking, gradient accumulation, and cross-language training (Portuguese, English). Cloud GPU orchestration on H100, A100 80GB, RTX 4090 via Vast.ai, RunPod, Lambda Labs. Cost-effective training pipelines (\$0.30-2.50/hour).

AI & Agent Systems: Python, PyTorch, Transformers, LangChain, OpenCV, Whisper, scikit-learn. YOLOv9/YOLOv11, video-based regression, time-series modeling. Built custom local LLM agent generators for document classification, smart contracts, and NLP-based workflows. Designed isolated agentic systems for offline usage (air-gapped / on-prem), including instant retraining modules. Multi-agent system coordination and embedding-based contextualization.

ML Infrastructure: Distributed training, asynchronous ML pipelines, model deployment optimization. Hugging Face ecosystem, Datasets, Accelerate, BitsAndBytes. Docker-containerized training environments, checkpoint management, evaluation metrics (BLEU, ROUGE, BERTScore). Real-time inference optimization and low-latency model serving.

Systems Engineering: Distributed system design, asynchronous pipelines, context-aware agent architecture. Docker, Kubernetes, serverless workflows, REST/gRPC API design. Real-time inference, low-latency system optimization. Experience with Rust-inspired system architecture and C++-style performance debugging.

Web & App Dev: React Native (5+ yrs), React (6+ yrs), Fabric & TurboModules. Next.js, TypeScript, Zustand, Redux, Deep Linking, OTA updates. Native integration: iOS (Swift), Android (Kotlin), biometric modules. WebSocket & GraphQL integrations for dynamic interfaces. Firebase, PostgreSQL, MongoDB, Express, Node.js.

DevOps & Cloud: GitHub Actions, Bitrise, Fastlane, Monorepo tooling. Cloud GPU management (Vast.ai, RunPod, Lambda Labs), cost optimization strategies. Docker-based isolated training environments. Local-only LLM training pipeline design (fully offline).

Notable Projects

Manual do Mundo LLM Fine-tuning (Production Content Generation)

- Fine-tuned Qwen2.5-7B model on 3,000+ Brazilian Portuguese educational video transcripts for Brazil's largest science YouTube channel (20M+ subscribers, 4.4B+ total views).
- Implemented LoRA adaptation with 4-bit quantization, achieving 27% loss reduction and generating production-ready video scripts in Manual do Mundo's authentic style.
- Deployed on RTX 4090 cloud infrastructure with memory optimization (7.8GB VRAM usage), reducing training costs to \$3 per run while processing 11MB+ datasets.
- Built smart chunking algorithms preserving narrative flow, quality scoring based on video performance, and production inference wrapper for content team integration.
- Enables 100x content scaling: from 2-3 videos/week to unlimited script generation for 15-25 minute educational content.

Cow Weight Estimation System (CV + Agent Workflow)

- Created a real-time video-based cattle weight predictor using OpenCV and PyTorch.
- Posture-aware, context-tracking model deployed on AMD hardware (non-CUDA).
- Integrated into WhatsApp for rural/farming contexts, enabling offline agent response.

Instant LLM Agent Generator (No Cloud Required)

- Tool that spawns intelligent AI agents from structured/unstructured docs.
- Use cases: contract analysis, paragraph classification, regulatory compliance support.
- Fully local, portable, and designed to operate in air-gapped infrastructures.

Document Processing Agents

- Built multi-phase agent pipelines: classification → contextualization → suggestion.
- Built on LangChain, Whisper, and custom embeddings.
- Integrated into Electron apps with local APIs.

Production Mobile Apps (React Native + New Architecture)

- Architected modular codebases with deep OTA control.
- Reduced load times by 35% through memory profiling and JS-native optimization.
- Built full test suites with Detox, RTL, and Jest.

Soft Skills & Culture Fit

- Exceptional async communication and remote-first practices.
- Strong sense of ownership, contributor mindset over manager hierarchy.
- Technical leadership, architectural planning, documentation best practices.
- Self-starter: thrives in startup environments with flat structure and high autonomy.
- Proven ability to deliver production AI systems under budget constraints with measurable business impact.

Eager to undertake a technical challenge to validate my skills in system design and implementation.

I work extremely well without any form of supervision. I like objective goals.

I welcome the opportunity to travel for onboarding to foster a strong team connection (I'm not AI).